Xi Li

Machine Learning Researcher — Biomedical AI and Computational Pathology

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Profile

Ph.D. researcher building multimodal AI for pathology, spatial genomics, and drug-target discovery. Research spans **STMoE**, **sMMC-22M**, and **TargetSage**, with experience translating research into deployed generative-AI products.

Education

University of California, Irvine	*Ph.D., Computer Science*	Sep 2023 – Present
Tsinghua University	*M.S., Computer Science and Technology*	Sep 2019 – Jun 2022
University of Science and Technology of China	*B.S., Information Security*	Sep 2015 – Jun 2019

Research Experience

University of California, Irvine **Irvine, CA**

Ph.D. Student Researcher *Sep 2023 – Present*

- **STMoE — histology-to-expression modeling.** Led development of a multi-scale mixture-of-experts model combining pathology-image embeddings, soft contrastive alignment, and organ-routed retrieval; restricted the molecular reference bank to training data.
- **Measured transfer performance:** on the matched human-lung cross-patient benchmark, gene-wise Pearson correlation was **0.2986 vs. 0.2248 for BLEEP** (**+0.0738 absolute**).
- **Data engineering and leakage control.** Built cell-aligned data and benchmarks for **CellST-24M** and **sMMC-22M**; tracked study/slide provenance, assay, and coordinate metadata, distinguishing native Xenium cells from derived Visium HD targets. The **expanded sMMC collection** contains **28.3M cell-aligned targets across 25 tissue categories**; the formal paper retains the frozen-release name sMMC-22M.
- **TargetSage — interpretable target prioritization (co-author).** Contributed to research combining biological-evidence profiling, LLM reasoning, and positive–unlabeled learning. The reported model improved **macro-average Adjusted F1 by 37% relative to the strongest of 10 baselines**, across 15 tasks and 19,032 genes.
- **iHAST — spatial super-resolution.** Developed a hybrid-attention model with wavelet features and spatial positional encoding. At **2× upscaling**, PSNR exceeded SwinIR by **2.37 / 2.89 / 4.39 dB** on Visium / Stereo-seq / Slide-seq, respectively (published benchmark).
- **Spine — spatial response modeling (first author).** Developed a formulation that separates tissue context from perturbation-conditioned response fields in spatial genomics; manuscript under review.

Multimedia Lab, Tsinghua University **Beijing, China**

Researcher, AI Broadcast Production *Oct 2021 – Sep 2022*

- Built an audio–visual cross-attention system for automated camera switching, crop selection, and virtual camera motion; **deployed at China's National Centre for the Performing Arts** for live orchestral production.

Technical Skills

Modeling Multimodal learning, transformers, mixture-of-experts, contrastive learning, LLM reasoning, positive–unlabeled models

Engineering Python, PyTorch, multi-GPU experimentation, image–omics alignment, dataset QC, grouped evaluation and reproducible benchmarking

Industry and Product Experience

Hantao Zhiying **China**
Chief Product Lead *2022 – 2023*
- Owned AI imaging product definition, user experience, and delivery coordination across research, engineering, and design.

ByteDance AI Lab **Beijing, China**
AI Product Manager Intern *May 2021 – Dec 2021*
- Incubated **Alchemy**, an internal image-generation platform; drove product scoping and internal launch for **Fanqie Novel book-cover creation**.
- Supported the **Chimera** digital-human project, owning automotive-client engagement and solution sales for **digital-human in-car assistants**.

ByteDance — Feishu/Lark **Beijing, China**
Data Product Intern *Jun 2020 – Nov 2020*
- Used behavior analytics and relevance analysis to inform enterprise document and knowledge-search product decisions.

Nolibox / Computational Aesthetics **Beijing, China**
Partner and Co-founder *Mar 2020 – Aug 2020*
- Developed automatic logo-generation and font-stroke generation tools for AI-assisted design.

Company milestone (post-tenure): Series A financing of tens of millions of RMB (2024).

Mentoring and Open-Source Leadership

Google Summer of Code (GSoC) **2025 & 2026**
Project Mentor — UC Santa Cruz Open Source Program Office
- **2026: Omni-ST** — mentored project design for instruction-driven multimodal modeling across histology, expression, spatial graphs, and text.
- **2025: Disentangled Generation and Editing of Pathology Images** — mentored generative-model and latent-editing design for controllable pathology images.

Selected Publications and Manuscripts

Full author lists and all papers: xenali96.github.io/publications.html.

Published / accepted

1. **Xi Li** et al. "STMoE: Multi-Scale Mixture-of-Experts for Single-Cell Gene Expression Prediction from Histology." **CIKM 2026**, accepted. *First author.*
2. **Xi Li** et al. "sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics." **NeurIPS 2026**, accepted poster. *First author.*
3. Z. Duan*, **Xi Li***, et al. "MUSE: A Multi-Slice Joint Analysis Method for Spatial Transcriptomics Experiments." **CIKM 2025**. **Co-first author.**
4. Z. Duan, T. Wu, **Xi Li**, et al. "TargetSage: Identifying Therapeutic Target Genes with Interpretable and Robust LLM Reasoning." **NeurIPS 2026**, accepted poster.
5. **Xi Li** et al. "iHAST: Integrating Hybrid Attention for Super-Resolution in Spatial Transcriptomics." **BMVC 2024**. *First author.*
6. Z. Duan, **Xi Li**, et al. "DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion." **ICIP 2025**.

Manuscripts

7. L. Lin, S. Bao, **Xi Li**, et al. "Spatially Indexed Single-Cell Morphology and Transcriptome Profiling by DBiC-seq." **Nature Communications**, under review.
8. **Xi Li** et al. "Context Is Not a Side Note: A Support-Calibrated Reliability Framework for Biomedical Foundation Models." **KDD 2027**, under review. *First author.*
9. **Xi Li** et al. "Spine: Disentangling Tissue Context and Perturbation-Conditioned Response Fields in Spatial Genomics." **KDD 2027**, under review. *First author.*
10. "CCR2 Deficiency Attenuates Flow-Induced Carotid Artery Remodeling." **Vascular Pharmacology**, under review. **Xi Li**: co-author.
11. F. Gao et al. "Cross-Species Multi-Omics Identifies a Conserved Fibrogenic Valvular Interstitial Cell Population across Mitral Valve Diseases." **Circulation**, submitted. **Xi Li**: co-author.